10/7/2026

Assignment No 01

Full-Stack Development Lab

Instructor: Sir Hairs

Submitted by: Wasif Ali

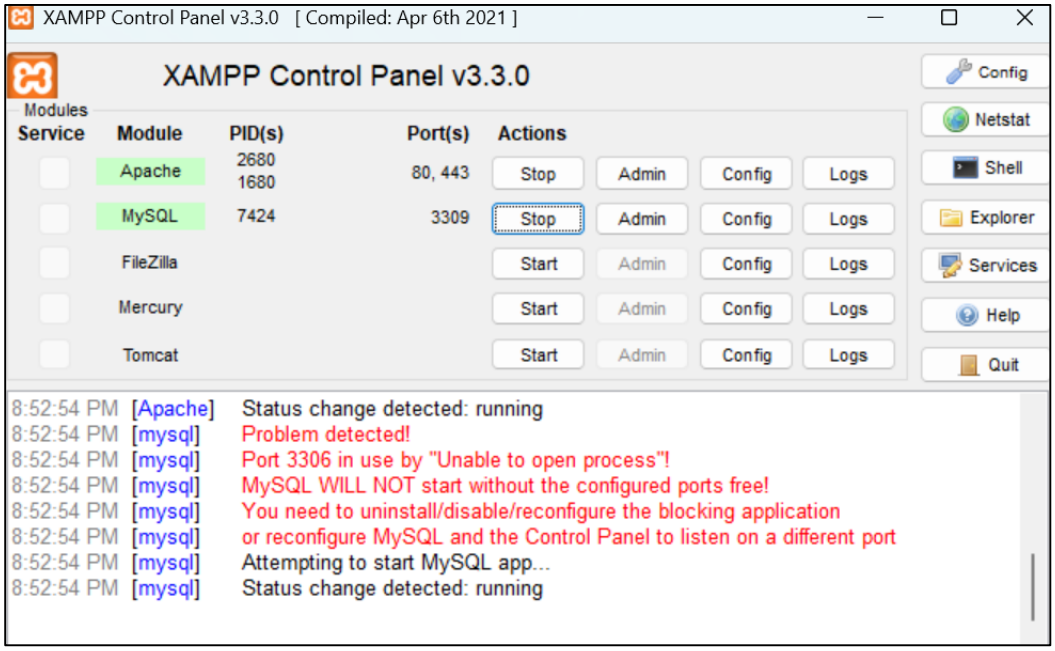
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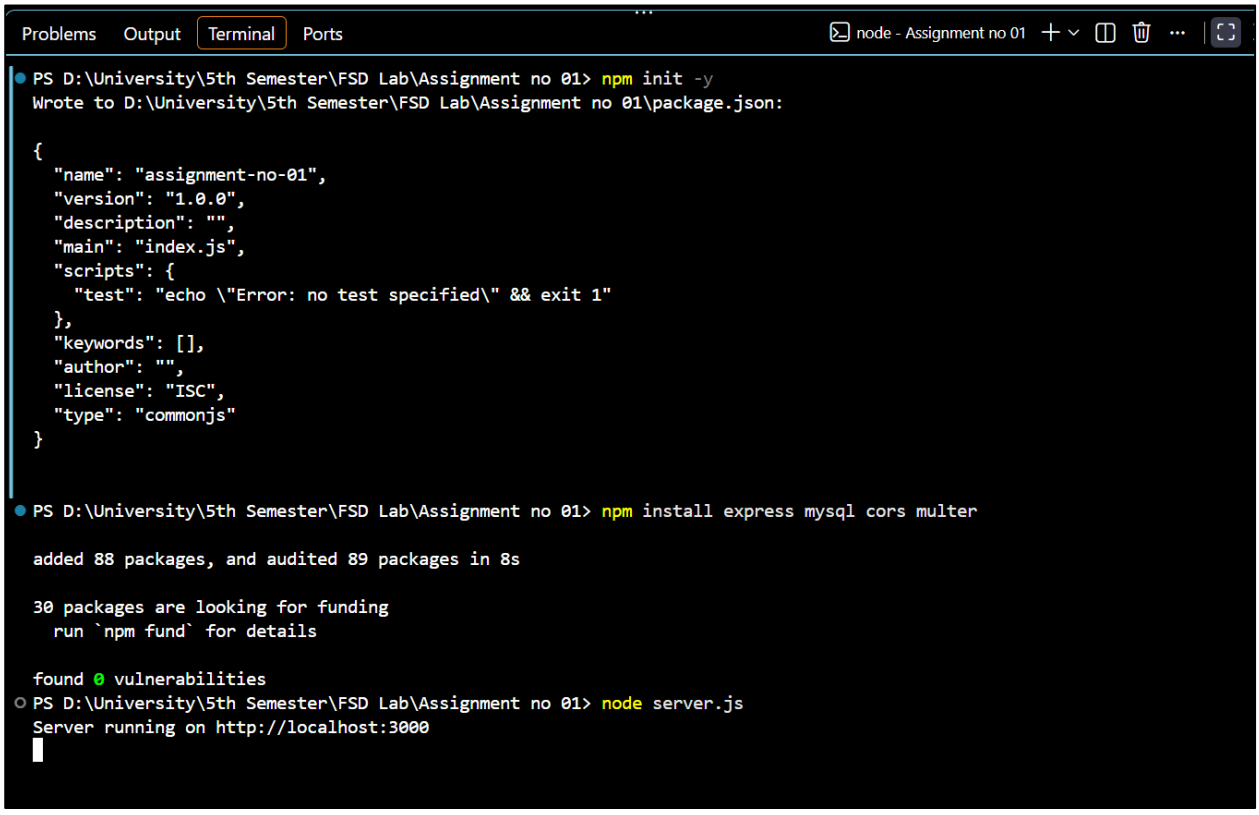


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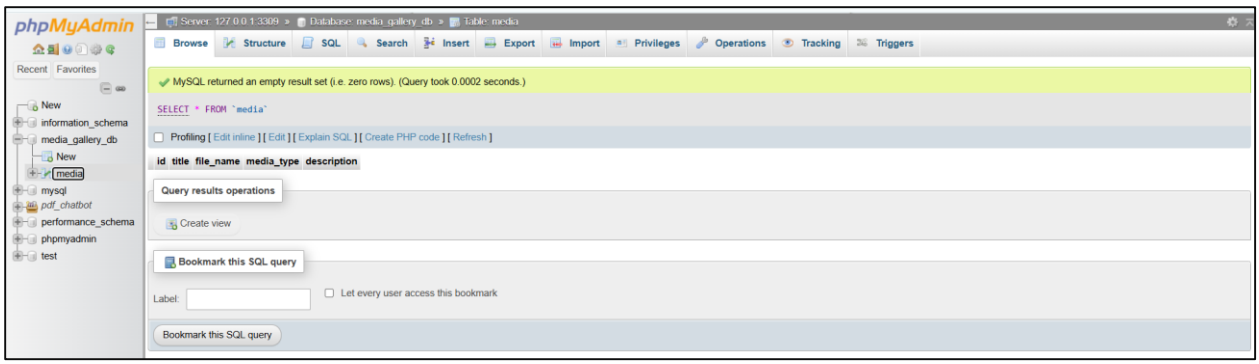
XAMPP Control Panel:



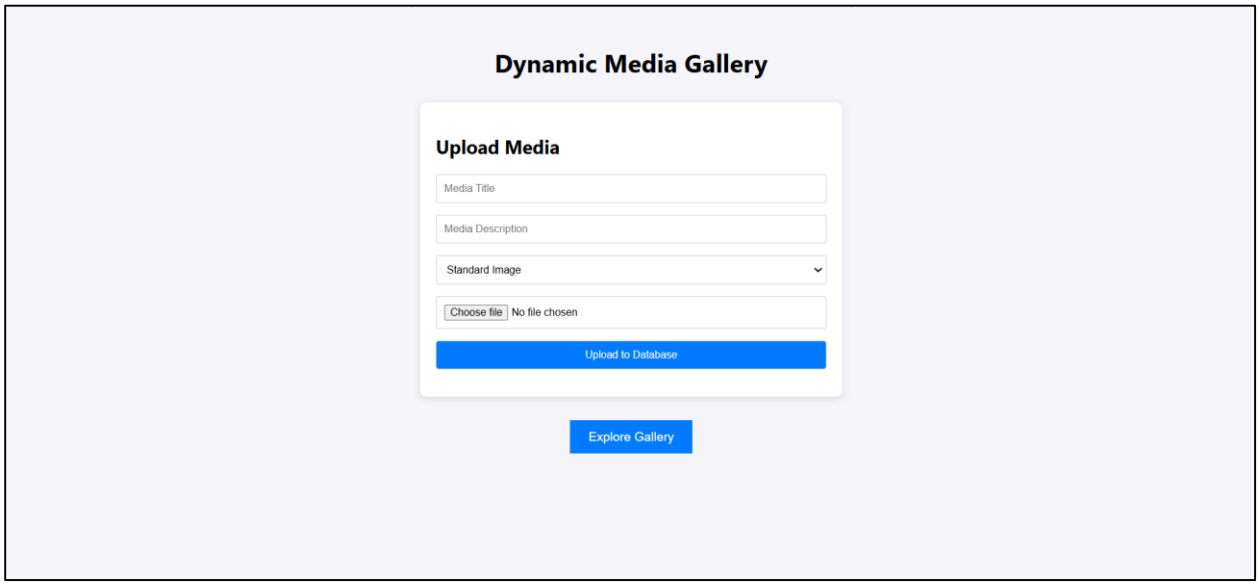
Server running

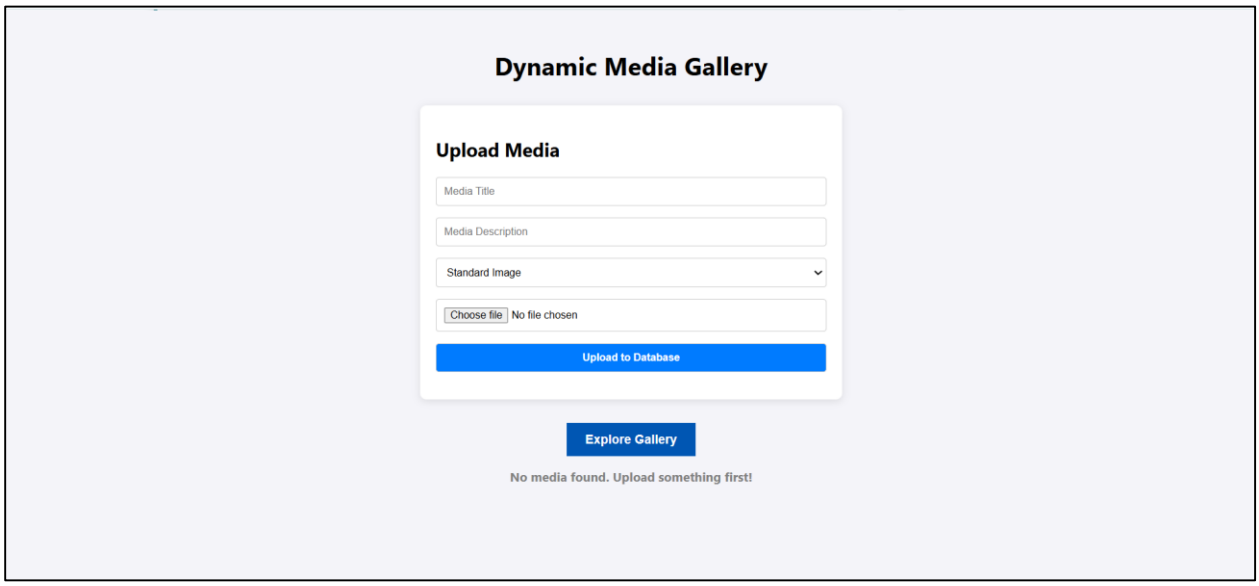


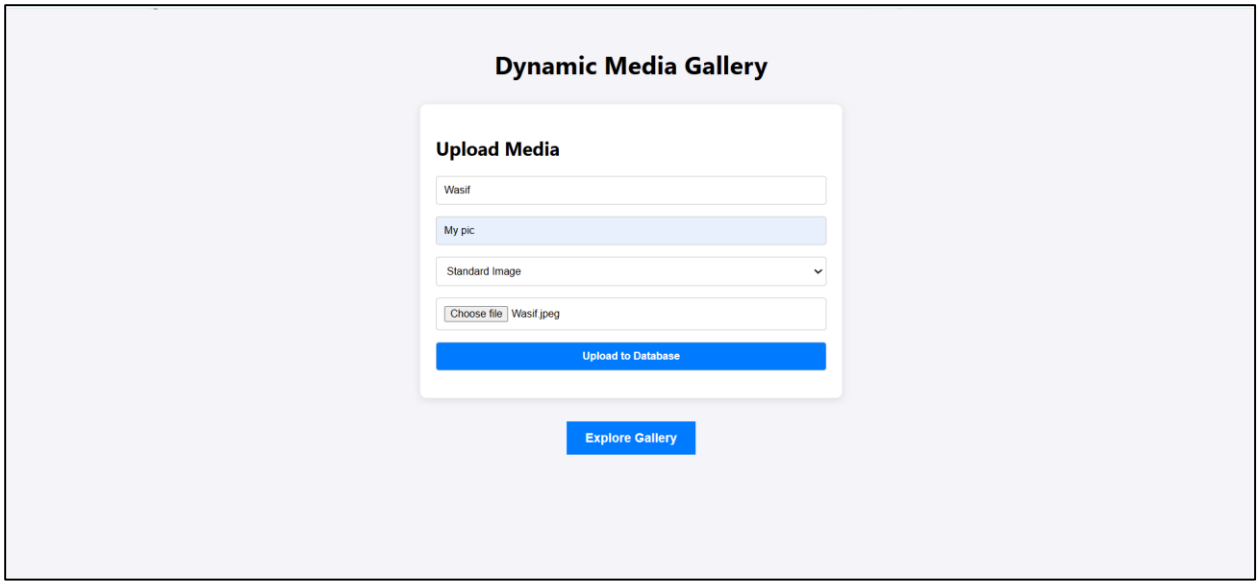
PhpMyAdmin



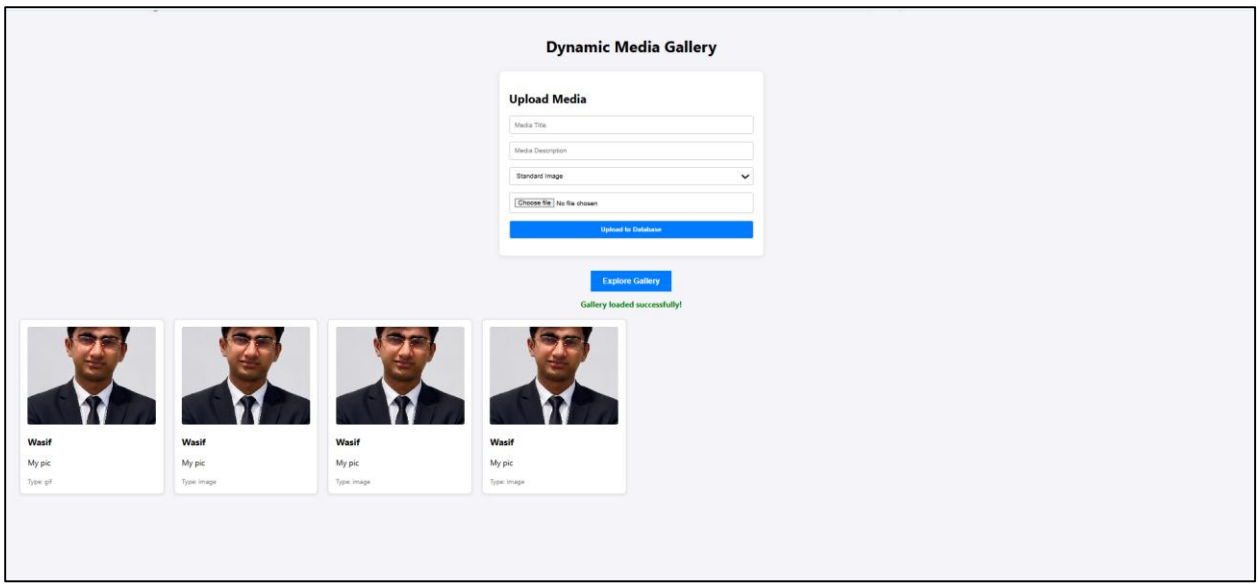
Frontend











Server.js

```
const express = require('express');
const mysql = require('mysql');
const cors = require('cors');
const multer = require('multer');
const path = require('path');

const app = express();
```

```
const port = 3000;

// Middleware
app.use(cors()); // Allows the front-end to communicate with this API
app.use(express.json());
app.use(express.urlencoded({ extended: true }));

// Expose the uploads folder so the front-end can display the images/GIFs
app.use('/uploads', express.static(path.join(__dirname, 'public/uploads')));

// 1. Database Connection Pool
const con = mysql.createPool({
    host: "localhost",
    user: "root",
    password: "",
    database: "media_gallery_db",
    port: 3309, //
    connectionLimit: 10
});

// 2. Configure Multer for File Uploads
const storage = multer.diskStorage({
    destination: function (req, file, cb) {
        cb(null, 'public/uploads/');
    },
    filename: function (req, file, cb) {
        // Generates a unique filename using the current timestamp
        cb(null, Date.now() + path.extname(file.originalname));
    }
});
const upload = multer({ storage: storage });

// 3. Upload Route (POST endpoint)
app.post('/upload', upload.single('mediaFile'), (req, res) => {
    // Extract text fields from the request body
    const { title, media_type, description } = req.body;

    // Extract the newly generated filename from multer
    const file_name = req.file.filename;

    // Insert the record into MySQL securely using placeholders
    const sql = "INSERT INTO media (title, file_name, media_type, description)
VALUES (?, ?, ?, ?)";
```

```
    con.query(sql, [title, file_name, media_type, description], (err, result) =>
{
        if (err) {
            console.error(err);
            return res.status(500).json({ message: "Database Error" });
        }
        res.status(200).json({ message: "Media uploaded and saved to database
successfully!" });
    });
});

// 4. GET Endpoint for jQuery AJAX
app.get('/media', (req, res) => {
    // Fetch all records, newest first
    const sql = "SELECT * FROM media ORDER BY id DESC";

    con.query(sql, function (err, result) {
        if (err) {
            console.error(err);
            return res.status(500).json({ message: "Database Error" });
        }
        // Returns the database records as a JSON array to the browser
        res.json(result);
    });
});

// Start the server
app.listen(port, () => {
    console.log(`Server running on http://localhost:${port}`);
});
```

Index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <title>Dynamic Media Gallery</title>

    <!-- Include jQuery CDN -->
    <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>

    <style>
        body {
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
```

```
        padding: 20px;
        background-color: #f4f4f9;
    }

    .header {
        text-align: center;
        margin-bottom: 30px;
    }

    .upload-section {
        background: white;
        padding: 20px;
        border-radius: 8px;
        box-shadow: 0 2px 10px rgba(0, 0, 0, 0.1);
        max-width: 500px;
        margin: 0 auto 30px;
    }

    .upload-section input,
    .upload-section select,
    .upload-section button {
        width: 100%;
        margin-bottom: 15px;
        padding: 10px;
        border: 1px solid #ccc;
        border-radius: 4px;
        box-sizing: border-box;
    }

    button {
        background-color: #007bff;
        color: white;
        border: none;
        cursor: pointer;
        transition: background-color 0.3s;
        padding: 10px;
        font-weight: bold;
    }

    button:hover {
        background-color: #0056b3;
    }

    .gallery-grid {
        display: grid;
```

```
            grid-template-columns: repeat(auto-fill, minmax(280px, 1fr));
            gap: 20px;
            margin-top: 20px;
        }

        .media-card {
            background: white;
            border: 1px solid #ddd;
            padding: 15px;
            border-radius: 8px;
            transition: transform 0.3s ease;
            box-shadow: 0 2px 5px rgba(0, 0, 0, 0.1);
        }

        .media-card:hover {
            transform: scale(1.05);
        }

        .media-card img {
            width: 100%;
            height: 200px;
            object-fit: cover;
            border-radius: 4px;
        }

        #status {
            text-align: center;
            margin: 15px 0;
            font-weight: bold;
        }
    </style>
</head>

<body>

    <div class="header">
        <h1>Dynamic Media Gallery</h1>
    </div>

    <div class="upload-section">
        <h2>Upload Media</h2>

        <form id="uploadForm" enctype="multipart/form-data">

            <input type="text" name="title" placeholder="Media Title" required>
```

```
            <input type="text" name="description" placeholder="Media Description"
required>

            <select name="media_type">
                <option value="image">Standard Image</option>
                <option value="gif">Animated GIF</option>
            </select>

            <input type="file" name="mediaFile" accept="image/*" required>

            <button type="submit">Upload to Database</button>

        </form>
    </div>

    <div style="text-align: center;">

        <button id="loadMedia" style="padding: 12px 24px; font-size: 16px;">
            Explore Gallery
        </button>

        <div id="status"></div>

    </div>

    <div id="gallery" class="gallery-grid"></div>


    <script>

        // Ensure DOM is ready before executing scripts
        $(document).ready(function () {

            // ---------------------------------------
            // 1. Handle File Upload via AJAX
            // ---------------------------------------

            $('#uploadForm').submit(function (e) {

                e.preventDefault();

                $('#status')
                    .html('Uploading...')
                    .css('color', 'blue');
```

```
        let formData = new FormData(this);

        $.ajax({

            url: 'http://localhost:3000/upload',

            type: 'POST',

            data: formData,

            contentType: false,

            processData: false,

            success: function (response) {

                $('#status')
                    .html(response.message)
                    .css('color', 'green');

                $('#uploadForm')[0].reset();
            },

            error: function (xhr) {

                $('#status')
                    .html('Upload failed. Check server connection.')
                    .css('color', 'red');
            }

        });

    });


    // ----------------------------------------
    // 2. Fetch and Display Gallery via AJAX
    // ----------------------------------------

    $('#loadMedia').click(function () {

        $('#status')
            .html('Loading gallery...')
            .css('color', 'blue');
```

```
                $.ajax({

                    url: 'http://localhost:3000/media',

                    type: 'GET',

                    dataType: 'json',

                    success: function (data) {

                        $('#status').html('');

                        $('#gallery').empty();


                        // Check if there is no media
                        if (data.length === 0) {

                            $('#status')
                                .html('No media found. Upload something first!')
                                .css('color', 'gray');

                            return;
                        }


                        // Display each media item
                        $.each(data, function (index, item) {

                            const card = `
                                <div class="media-card">

                                    <img
                                        src="http://localhost:3000/uploads/${item
.file_name}"

                                        alt="${item.title}"
                                    >

                                    <h3>${item.title}</h3>

                                    <p>${item.description}</p>

                                    <small style="color: gray;">
                                        Type: ${item.media_type}
```

```
                                    </small>

                                </div>
                            `;

                            $('#gallery').append(card);

                        });


                        $('#status')
                            .html('Gallery loaded successfully!')
                            .css('color', 'green');

                    },


                    error: function (xhr, status, error) {

                        $('#status')
                            .html('Error loading media! Ensure Node.js server is
running.')
                            .css('color', 'red');

                        console.log(xhr.responseText);
                    }

                });

            });

        });

    </script>

</body>

</html>
```